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Machine learning or deep learning?

Five questions to answer before choosing an approach for a new AI project. Use it in a project kickoff to match the method to the data and to the business need, not to the hype.

Ask your team	Points toward traditional ML	Points toward deep learning
1 **What does the data look like?**	Rows and columns: customer records, claims, prices, transactions	Images, audio, video, free text or raw sensor streams
2 **Do we already know which features matter?**	Yes. Domain experts can name them, such as tenure, contract type or monthly charges	No. The signal sits in raw patterns that no one can fully describe by hand
3 **How much labeled data do we have?**	A moderate amount, typical of business records	A very large amount, or a pretrained model we can adapt
4 **Must people understand each prediction?**	Yes. Regulated, financial or clinical decisions need explainable drivers	Accuracy comes first, and results can be validated in other ways
5 **What can we spend on compute and upkeep?**	Standard hardware, low cost, frequent retraining	GPU training and specialized engineering support are available

Mostly left column
Start with traditional ML: logistic regression, tree-based models or support vector machines.

Mostly right column
Use deep learning: convolutional networks for images, transformer models for text.

A mix of both
Build a simple, explainable baseline first. Move to deep learning only if it clearly beats the baseline.

Two worked examples

Traditional ML: support vector machine

Predicting customer churn

Telecom churn data is tabular, and its predictive features are already known to the business. An SVM separates likely leavers from stayers on moderate data, trains cheaply and supports explanations that retention teams can act on. A deep network would add cost and opacity without a meaningful gain.

Deep learning: convolutional neural network

Recognizing faces with Face ID

Depth maps and infrared images are unstructured and high-dimensional. Layered networks learn edges, then facial parts, then whole faces, and stay reliable across lighting, angles and glasses. Hand-crafted features such as eye distance are too brittle for a security system.

In healthcare analytics: pharmacy pricing and claims data are tabular, so traditional ML is usually the right starting point and keeps decisions explainable to stakeholders. Medical images are unstructured, so deep learning is usually the stronger choice.

Sources: Grinsztajn, Oyallon & Varoquaux (2022), tree-based models on tabular data, NeurIPS 35; LeCun, Bengio & Hinton (2015), Deep learning, *Nature* 521; Vafeiadis et al. (2015), churn prediction techniques, *Simulation Modelling Practice and Theory* 55; Apple Inc., About Face ID advanced technology.